

Chapter 1

Section 2 - Structures: Sets, Functions, Sequences, Sums, and Matrices

1.1 Section 2.1 - The Joy of Sets

Definition 1 (Sets and Elements)

A **set** is an unordered collection of objects called **elements**

$$\mathbb{N} = \{1, 2, 3, 4, \dots\} \quad (\text{Natural Numbers})$$

$$\text{Whole Numbers} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{0, -1, 1, -2, 2, -3, 3, \dots\}$$

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$$

$$\mathbb{R} = \text{limit of all convergent sequences of rational numbers}$$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$$

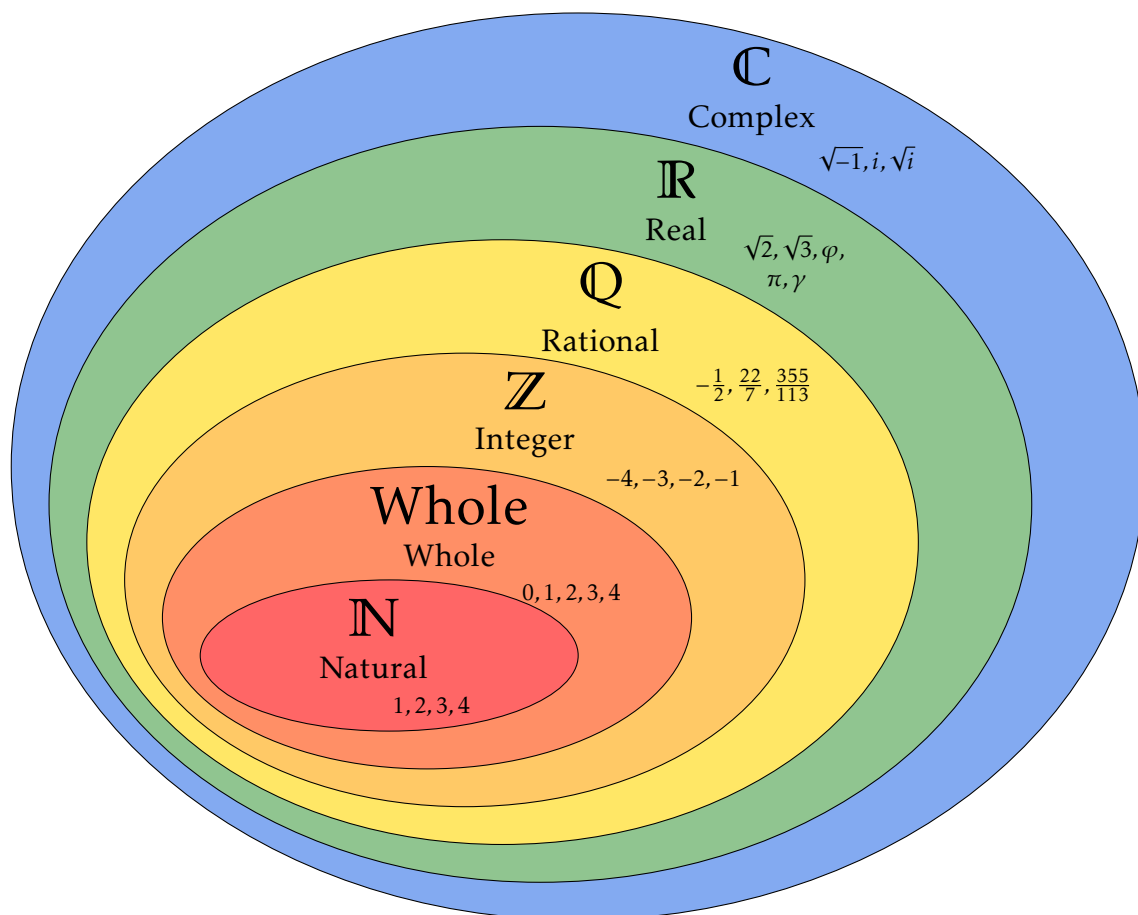
Note

This notation for $\mathbb{Q}$ is called **Set Builder Notation**

It's read as "Q is the set $\frac{a}{b}$ such that $a, b \in \mathbb{Z}, b \neq 0$ "

Note

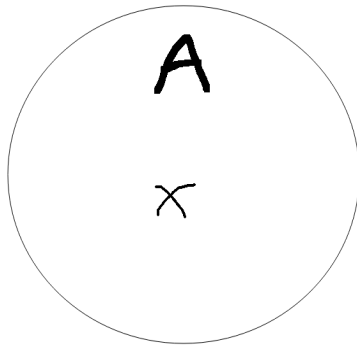
Notice that every natural number is a whole number. So we say "the set of natural numbers is a **subset** of the set of whole numbers"



Definition 2 (Elements in Sets)

If x is an element of the set A , then we write

$$x \in A$$



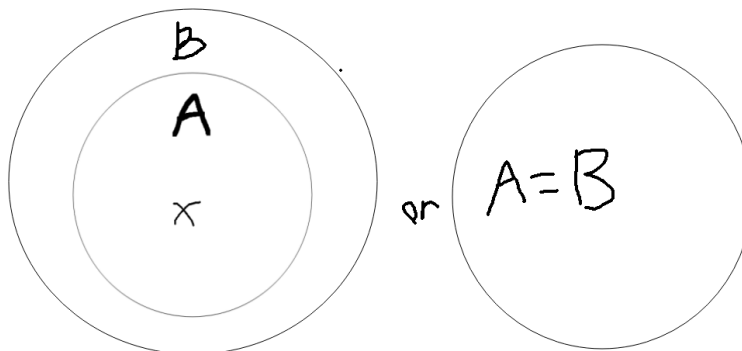
Definition 3 (Empty Set)

The set containing no elements is called the **empty set**, and is denoted by $\{\}$ or $\emptyset$

Given sets A and B such that for all $x \in A$ we have $x \in B$, then we say **A is a subset of B** and we write

$$A \subseteq B$$

Which can mean two things:



Definition 4 (Proper Sets)

If there exists $x \in B$ with $x \notin A$, then we say that A is a **proper subset** of B and we write

$$A \subsetneq B$$

Definition 5 (Cardinality)

Given a set A , the **cardinality** of a set is the number of elements in the set A , denoted by

$$|A|$$

Definition 6 (Power Set)

The **power set** of a given set S is the set of all subsets of S , denoted by

$$\mathcal{P}(S)$$

$$|\mathcal{P}(S)| = 2^{|S|}$$

Note

This includes the empty set, $\{\}$

Definition 7 (Cartesian Product)

Given sets A, B , the **cartesian product** is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$$|A \times B| = |A| \cdot |B|$$

$$A^2 = A \times A$$

Example 1

Let $A = \{1, 2, 3\}, B = \{1, 2\}$

(a) Is $A \subseteq B$?

Solution:

No since $3 \in A, 3 \notin B$

(b) Is $B \in A$?

Solution:

Yes since for all $x \in B, x \in A$

(c) What is the power set of B ?

Solution:

$$\mathcal{P} = \{\{\}, \{1\}, \{2\}, \{1, 2\}\}$$

(d) What is the power set of A ?

Solution:

$$\mathcal{P}(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note

If A is a finite set (ie $|A|$ is finite), then $|\mathcal{P}(A)| = 2^{|A|}$

(e) Determine $A \times B$

Solution:

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$$B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A| = |A| \cdot |B|$$

1.2 Section 2.2 - Set Operations

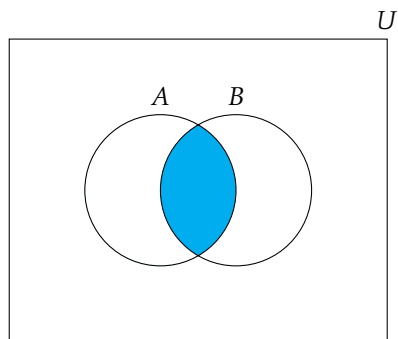
Let's consider a venn diagram which is a way of visualizing sets

Let U represent the universal set

Definition 8 (Intersection)

Elements that are in **both** A and B is called the intersection of A and B

$$A \cap B$$



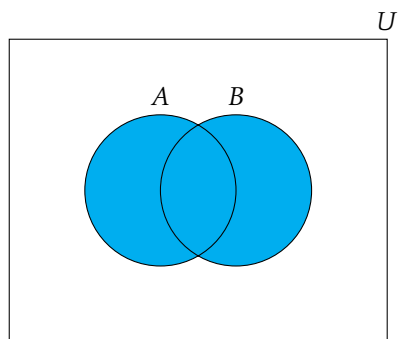
Definition 9 (Union)

Elements that are in **either** A or B is called the **union** of A and B

$$A \cup B$$

Also

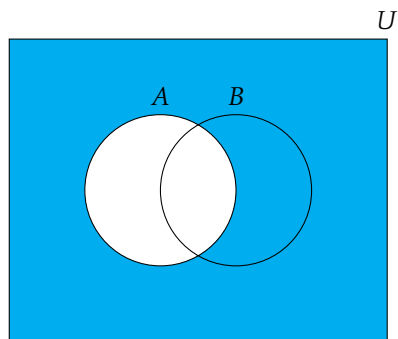
$$|A \cup B| = |A| + |B| - |A \cap B|$$



Definition 10 (Compliment)

Elements that just aren't in A is called the **compliment** of A

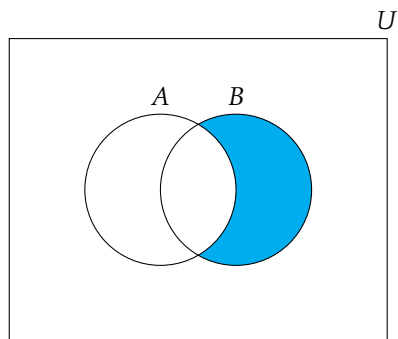
$$\bar{A}$$



Definition 11 (Difference)

Elements in B but not in A is called the **difference**

$$B - A$$



With sets, we have 3 operations:

1. Compliment $\bar{A}$ (analogous to $\neg$)
2. Union $A \cup B$ (analogous to $\vee$)
3. Intersection $A \cap B$ (analogous to $\wedge$)

Example 2

Let

$$U = \{x \in \mathbb{Z} \mid 0 \leq x \leq 20\}$$

$$A = \{x \in U \mid x \text{ is even}\}$$

$$B = \{x \in U \mid x \text{ is divisible by 3}\}$$

Determine the following

1. $A = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$
2. $B = \{0, 3, 6, 9, 12, 15, 18\}$
3. $A \cap B = \{0, 6, 12, 18\}$ (what they have in common)
4. $A \cup B = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$ (whats in either one)
5. $\bar{A} = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ (whats not in A)
6. $B - A = \{3, 9, 15\}$ (what's in B thats not in A)
7. $|A| = 11$ (number of elements in A)
8. $|B| = 7$ (number of elements in B)
9. $|A \cup B| = 14$
10. $|A \cap B| = 4$ (number of elements that A and B have in common)
11. $|A| + |B| - |A \cap B| = 11 + 7 - 4 = 14 = |A \cup B|$

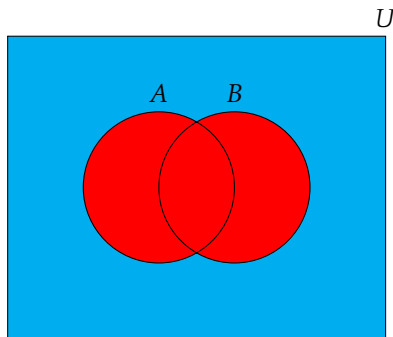
But how do these operations interact together?

Example 3

- $\overline{A \cup B}$

$$\text{Red} = A \cup B$$

$$\text{Blue} = \overline{A \cup B}$$

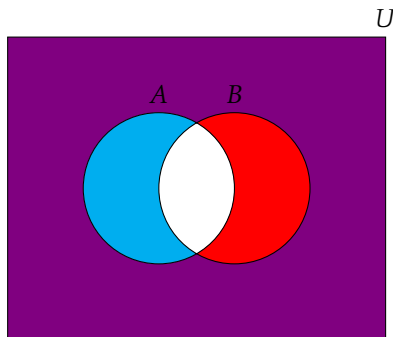


- $\bar{A} \cap \bar{B}$

$$\text{Red} = \bar{A}$$

$$\text{Blue} = \bar{B}$$

$$\text{Purple} = \bar{A} \cap \bar{B}$$



$$\therefore \overline{A \cup B} = \overline{A} \cap \overline{B}$$

1. **DeMorgan's Laws**

- $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2. **Distributive Laws**

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

3. **Complementation Law**

- $\overline{\overline{A}} = A$

4. **Commutative Law**

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

1.3 Section 2.3 - Functions

Definition 12

Let A and B be sets

A **relation** from A to B is a subset of $A \times B$

Example 4

Let $A = \{1, 4, 9\}$ and $B = \{\pm 1, \pm 2, \pm 3\}$

1. $|A \times B| = 3 \cdot 6 = 18$
2. Let R be the relation from A to B defined by

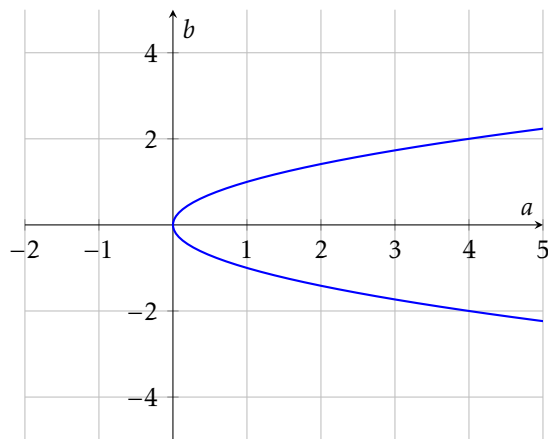
$$R = \{(a, b) \mid a \in A, b \in B, a = b^2\}$$

$$R = \{(1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3)\}$$

Note

R is a relation but not a function

Why?



This graph of the relation fails the vertical line test, hence it is not a function

There exists an element of A that corresponds to more than one element of B

a	b
1	-1
1	1
4	-2
4	2
9	-3
9	3

Every function is a relation but not all relations are functions

Definition 13 (Function)

Let A and B be sets

A **function** f from A to B , denoted as

$$f : A \rightarrow B$$

is a relation from A to B such that

$$\forall a \in A \exists! b \in B, f(a) = b$$

$\exists!$ means "there exists a **unique**..."

Definition 14 (Domain and Codomain)

We call the set A the **domain** of f and the set B the **codomain**.

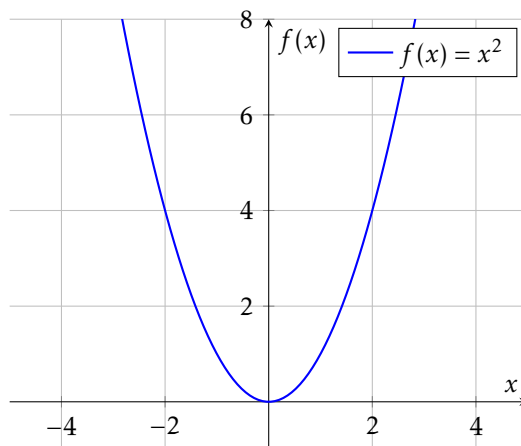
The **range** or **image** of f is the set

$$\text{Im}(f) = \{b \in B \mid \exists a \in A, f(a) = b\}$$

Example 5

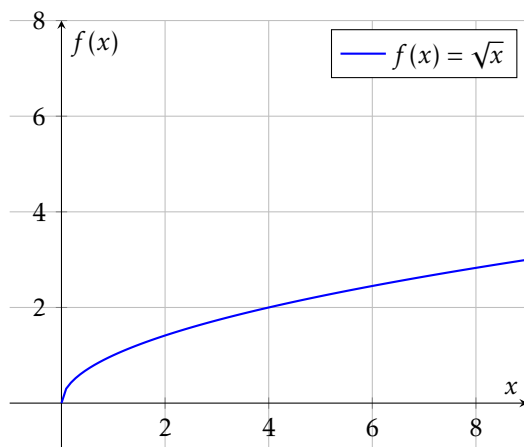
Determine if the following are functions

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$



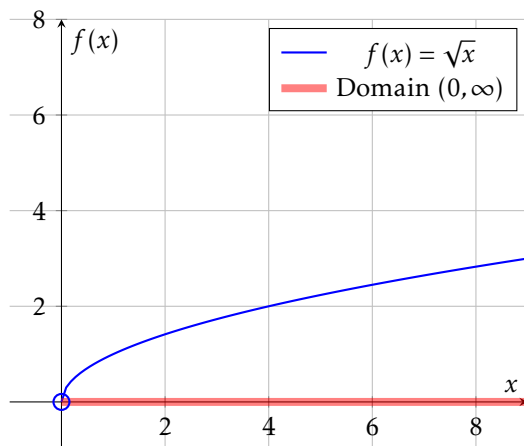
Domain: x axis
 Codomain: y axis
 $\text{Im}(f) = [0, \infty]$
 $\therefore$ Yes

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



No since not all inputs are valid

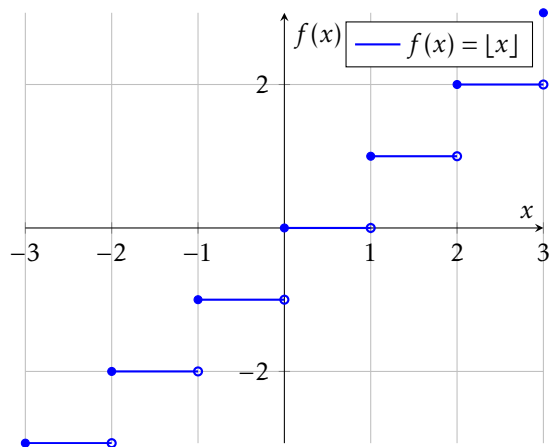
3. $f : \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



$$\text{Im}(f) = (0, \infty)$$

Yes since all the inputs have a unique output (the origin is excluded since the domain is $\mathbb{R}^+$)

4. $f : \mathbb{R} \rightarrow \mathbb{Z}, f(x) = \lfloor x \rfloor$ (floor function, aka greatest integer function)

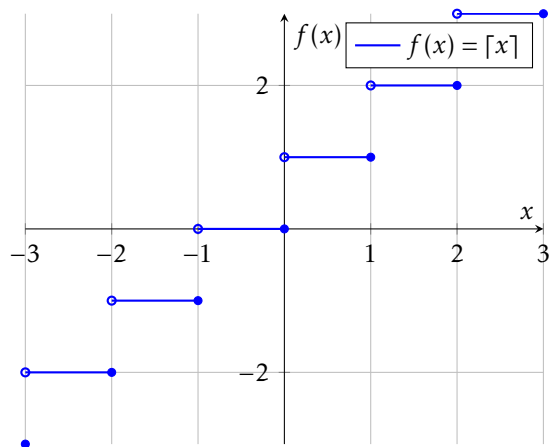


x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	0
0.99	0
1.01	1

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

5. $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lceil x \rceil$ (ceiling function, aka least integer function)



x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	1
0.99	1
1.01	2

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

Definition 15 (Surjective)

Let $f : A \rightarrow B$ be a function in the set A and B

We say f is **onto** or **surjective** if

$$\text{Im}(f) = B$$

or

$$\forall b \in B \exists a \in A, f(a) = b$$

or

All of codomain is covered by the function

Definition 16 (Injective)

We say f is **one to one** or **injective** if

$$\forall b \in B, \text{ the equation } f(a) = b \text{ has at most one solution}$$

Another way of thinking of this is:

Injective if it passes the horizontal line test

or

there is a unique b for all a

or

no two inputs have the same output

or

$$f(a) = f(b) \implies a = b$$

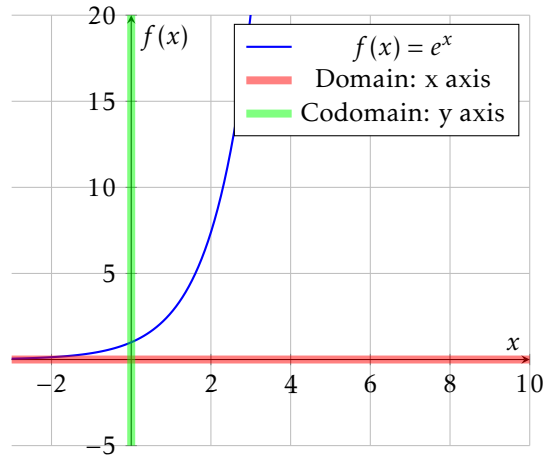
Definition 17 (Bijective)

If f is both **surjective** and **injective**, then we say f is **bijective**

Example 6

Determine if the following functions are injective, surjective, and / or bijective

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

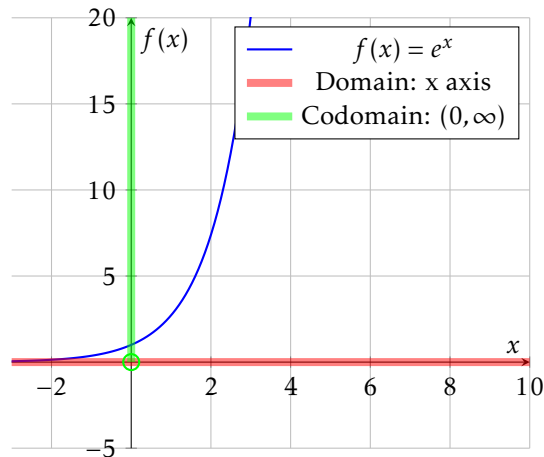


Injective since all x values have a valid output

Not surjective since not all values on the y axis have a corresponding input value

Not bijective since it's not surjective

2. $f : \mathbb{R} \rightarrow \mathbb{R}^+, f(x) = e^x$



Injective since all x values have a valid output

Surjective since all values on the y axis have a corresponding input value

Bijjective since it's both surjective and injective

There's more examples, write them down when you get the chance

Definition 18 (Composition)

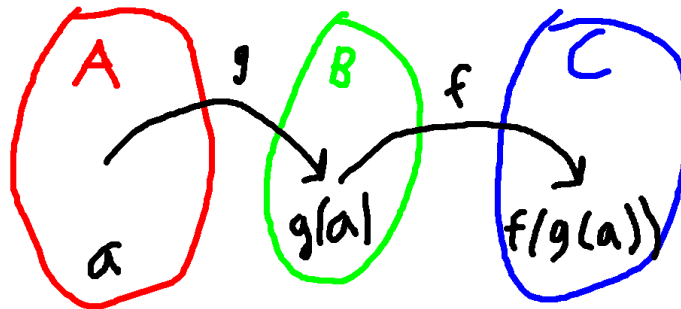
Given the functions $f : B \rightarrow C$ and $g : A \rightarrow B$

Where A, B, C are sets, then we define

$$f \circ g : A \rightarrow C \text{ by}$$

$$(f \circ g)(a) = f(g(a))$$

called the **composition** of f and g



Example 7 (Examples of Compositions)

Let $f(x) = x - 3$, $g(x) = x^3 + 9x^2 + 27x + 27 = (x + 3)^3$

Compute the following

1. $(f \circ g)(x)$

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) \\
 &= (x^3 + 9x^2 + 27x + 27) - 3 \\
 &= x^3 + 9x^2 + 27x + 24
 \end{aligned}$$

2. $(g \circ f)(x)$

$$\begin{aligned}
 (g \circ f)(x) &= g(f(x)) \\
 &= (x - 3)^3 + 9(x - 3)^2 + 27(x - 3) + 27 \\
 &= [(x - 3) + 3]^3 \\
 &= x^3
 \end{aligned}$$

1.4 Section 2.4 - Sequences and Series

Definition 19 (Sequence, Explicit Formula / nth term)

A **sequence** is an ordered list of numbers

$$a_1, a_2, a_3, \dots$$

Equivalently, a sequence is a function $f : \mathbb{Z}^+ \rightarrow \mathbb{R}$

We call f the **explicit formula** or **nth term** of a sequence

Example 8

Determine the next 3 terms of the following sequences and the explicit formula

$$1. \quad 5 \xrightarrow{+7}, 12 \xrightarrow{+7}, 19 \xrightarrow{+7}, 26 \xrightarrow{+7}, \dots, 33 \xrightarrow{+7}, 44 \xrightarrow{+7}$$

Solution:

$$\begin{aligned}
 a_n &= 7n + b \\
 5 = a_1 &= 7(1) + b \\
 b &= -2 \\
 \boxed{a_n} &= \boxed{7n - 2}
 \end{aligned}$$

$$2. \quad 7 \underbrace{\longrightarrow}_{\cdot 3}, 21 \underbrace{\longrightarrow}_{\cdot 3}, 63 \underbrace{\longrightarrow}_{\cdot 3}, 189 \underbrace{\longrightarrow}_{\cdot 3}, \dots, 567 \underbrace{\longrightarrow}_{\cdot 3}, 1701 \underbrace{\longrightarrow}_{\cdot 3}, 5103 \underbrace{\longrightarrow}_{\cdot 3}$$

Solution:

$$\begin{aligned}
 a_n &= c3^n \\
 a_1 &= 7 = c3^1 \\
 c &= \frac{7}{3} \\
 a_n &= \frac{7}{3}3^n \\
 \boxed{a_n} &= \boxed{7 \cdot 3^{n-1}}
 \end{aligned}$$

$$3. \quad 1, 2, 6, 24, 120, 720, 5040$$

$\boxed{a_n = n!}$, read as "n factorial"

$$0! = 1$$

$$4. \quad 1, 1, \underbrace{2}_{1+1}, \underbrace{3}_{2+1}, \underbrace{5}_{3+2}, \underbrace{8}_{5+3}, \underbrace{13}_{8+5}, \underbrace{21}_{13+8},$$

Solution:

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right] \quad (\text{We'll derive this later})$$

Another solution is

$$\begin{aligned}
 a_1 &= 1 \\
 a_2 &= 2 \\
 a_n &= a_{n-1} + a_{n-2}, n \geq 3
 \end{aligned}$$

Definition 20

A **recursive** sequence is defined in terms of the previous terms of the sequence

Example 9

Here are some examples of recursive sequences

$$1. \ a_1 = 2, a_2 = 3, a_n + 1 = 5a_n - 6a_{n-1}, n \geq 2$$

$$\begin{aligned}
 n = 2, a_3 &= 5a_2 - 6a_1 = 5 \cdot 3 - 6 \cdot 2 = 15 - 12 = 3 \\
 n = 3, a_4 &= 5a_3 - 6a_2 = 5 \cdot 3 - 6 \cdot 3 = 15 - 18 = -3 \\
 n = 4, a_5 &= 5a_4 - 6a_3 = 5 \cdot (-3) - 6 \cdot 3 = -15 - 18 = -33
 \end{aligned}$$

$$2. \ a_1 = 1, a_2 = 2, a_3 = 4, a_n = a_{n-1} + a_{n-2} - n - 3, n \geq 4$$

$$\begin{aligned}
 a_4 &= 4 + 2 - 1 = 5 \\
 a_5 &= 5 + 4 - 2 = 7 \\
 a_6 &= 7 + 5 - 4 = 8
 \end{aligned}$$

Definition 21 (Series)

A **series** is the sum of an ordered list of numbers

$$a_1 + a_2 + a_3 + \dots$$

$$\sum_{k=1}^{\infty} a_k$$

Where:

k = Index
 1 = Starting value
 ∞ = Ending value
 $a_k = k^{\text{th}}$ term

Example 10

Find the sum of the series

$$1 + 2 + 3 + \dots + 100$$

Solution:

$$\begin{aligned}
 &1 + 2 + 3 + \dots + 98 + 99 + 100 \\
 &(1 + 100) + (2 + 99) + (3 + 98) + \dots \\
 &(101) + (101) + (101) + \dots
 \end{aligned}$$

Since there's 100 things we need to add, there's 50 pairs

So

$$\begin{aligned}
 \text{sum} &= 101(50) \\
 &= 5050
 \end{aligned}$$

Example 11

But what about in general?

$$1 + 2 + 3 + \dots + n = ?$$

We have two cases:

1. When n is even
2. When n is odd

If we assume n is even, then

$$\boxed{1 + 2 + 3 + \dots + n = (n + 1) \cdot \frac{n}{2}} \quad (1.1)$$

Since $(n + 1)$ gives the value of each pair and $\frac{n}{2}$ gives the number of pairs

If we assume n is odd, then

$$\begin{aligned} [1 + 2 + 3 + \dots + (n - 1)] + n &= n \cdot \frac{n - 1}{2} \\ &= \frac{n^2 + n}{2} \\ &= (n + 1) \cdot \frac{n}{2} \end{aligned}$$

since n is the value of each pair $(n - 1 + 1)$ and $\frac{n-1}{2}$ is the number of pairs

Another way to think about it is to let S represent the sum

$$\begin{array}{r} S = 1 + 2 + 3 + \dots + (n - 2) + (n - 1) + n \\ S = n + (n - 1) + (n - 2) + \dots + 3 + 2 + 1 \quad \text{(Switch Order)} \\ \hline 2S = (n + 1) + (n + 1) + (n + 1) + \dots + (n + 1) + (n + 1) + (n + 1) \quad \text{(Add)} \\ \underbrace{\hspace{10em}}_n \\ \boxed{S = (n + 1) \cdot \frac{n}{2}} \end{array}$$

Writing this in summation notation

$$\sum_{k=1}^n \frac{n(n + 1)}{2}$$

To Review...

1.

$$\sum_{k=1}^n c = \underbrace{c + c + \dots + c}_n = \boxed{nc} \quad \text{(North Carolina Rule)}$$

2.

$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \boxed{(n+1)\frac{n}{2}} \quad (\text{Pairing Rule})$$

3.

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \boxed{\frac{n(n+1)(2n+1)}{6}} \quad (\text{Prove Later})$$

4.

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \boxed{\left[\frac{n(n+1)}{2}\right]^2} \quad (\text{Prove Later})$$

You **MUST** know these 4! They will be on exams!

Example 12

Compute the Following

1.

$$\sum_{k=1}^6 k = \frac{6}{2}(6+1) = \boxed{21} \quad (\text{Use formula})$$

2.

$$\begin{aligned} \sum_{k=1}^6 2k &= (2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6) \\ &= 2(1 + 2 + 3 + 4 + 5 + 6) \\ &= 2 \sum_{k=1}^6 k \\ &= 2 \cdot 21 \\ &= \boxed{42} \end{aligned}$$

Note

This means that we can **factor** in sums!

3.

$$\sum_{k=1}^6 2 = 6 \cdot 2 = \boxed{12} \quad (\text{Use formula})$$

4.

$$\begin{aligned} \sum_{k=1}^6 (k+2) &= (1+2)(2+2) + (3+2) + (4+2) + (5+2) + (6+2) \\ &= (1+2+3+4+5+6) + (2+2+2+2+2+2) \\ &= \sum_{k=1}^6 k + \sum_{k=1}^6 2 \\ &= 21 + 12 \\ &= \boxed{33} \end{aligned}$$

Note

This means that in general,

$$\begin{aligned} \sum_{k=1}^n c a_k &= c \sum_{k=1}^n a_k && (\text{Factor out constants}) \\ \sum_{k=1}^n (a_k + b_k) &= \sum_{k=1}^n a_k + \sum_{k=1}^n b_k && (\text{Split}) \end{aligned}$$

5.

$$\begin{aligned} \sum_{k=1}^{200} (k+1)^2 &= \sum_{k=1}^{200} (k^2 + 2k + 1) && (\text{Expand}) \\ &= \sum_{k=1}^{200} k^2 + 2 \sum_{k=1}^{200} k + \sum_{k=1}^{200} 1 && (\text{Split up}) \\ &= \frac{200(201(401))}{6} + 2 \cdot \frac{200(201)}{2} + 200 \cdot 1 && (\text{Use formulas}) \\ &= \boxed{2727100} && (\text{Use Calculator}) \end{aligned}$$

We could also do

$$\begin{aligned}
 \sum_{k=1}^{200} (k+1)^2 &= 2^2 + 3^2 + 4^2 + \dots + 201^2 \\
 &= (1^2 + 2^2 + 3^2 + 4^2 + \dots + 201^2) - 1^2 \\
 &= \frac{201(202)(403)}{2} - 1 \\
 &= 2727100 \quad \text{(Use Calculator)}
 \end{aligned}$$

Theorem 1 (Constant Shift)

This means that when you have a **constant shift** like $(k+1)$ is a shift from k , then

$$\sum_{k=1}^n f(k+a) = \sum_{k=1}^{n+a} f(k) - \sum_{k=1}^a f(k)$$

or

$$\sum_{k=m}^n a_k = \sum_{k=1}^n a_k - \sum_{k=1}^{m-1} a_k$$

Where $f(k)$ is a function like k^2 that has it's own formula for the sum like

$$\sum_{k=1}^n f(k) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Using this on the other example,

$$\begin{aligned}
 \sum_{k=1}^{200} (k+1)^2 &= \sum_{k=1}^{201} k^2 - \sum_{k=1}^1 k^2 \\
 &= \frac{201(202)(403)}{2} - \frac{1(2)(1)}{2} \\
 &= 2727101 - 1 \\
 &= 2727100 \quad \text{(Use Calculator)}
 \end{aligned}$$

6.

$$\sum_{k=1}^{10} 2^k = S$$

$$S = 2^1 + 2^2 + 2^3 + \dots + 2^9 + 2^{10}$$

$$2S = 2^2 + 2^3 + 2^4 + \dots + 2^{10} + 2^{11}$$

(Subtract)

$$S - 2S = 2^1 - 2^{11}$$

$$S(1 - 2) = 2^1 - 2^{11}$$

$$S = \frac{2 - 2^{11}}{1 - 2} = \frac{2 - 2048}{-1}$$

$$= 2046$$

Note

This means that

$$S = \sum_{k=0}^n r^k \quad (\text{When } k \text{ starts at } 0)$$

$$S = 1 + r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$rS = 0 + r^1 + r^2 + r^3 + \dots + r^{n-1} + r^n + r^{n+1}$$

(Subtract)

$$S - rS = 1 - r^{n+1}$$

$$S(1 - r) = 1 - r^{n+1}$$

$$\boxed{S = \frac{1 - r^{n+1}}{1 - r}}$$

$$S = \sum_{k=1}^n r^k \quad (\text{When } k \text{ starts at } 1)$$

$$S = r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$= r(1 + r^1 + r^2 + \dots + r^{n-2} + r^{n-1})$$

$$= r \sum_{k=1}^{n-1} r^k$$

$$= r \cdot \frac{1 - r^{n-1+1}}{1 - r}$$

$$\boxed{= r \cdot \frac{1 - r^n}{1 - r}}$$

In general,

$$\boxed{\sum_{k=\text{start}}^n r^k = \frac{r^{\text{start}} - r^{n+1}}{1 - r}}$$

Note

This works for $r \in \mathbb{R}, (-\infty, \infty)$, including **negative** and **fractions**!

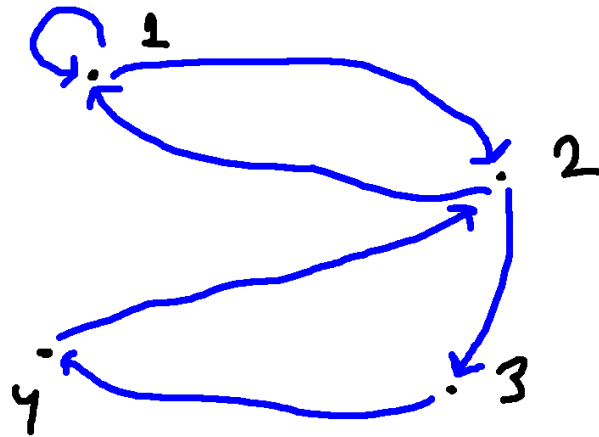
This can also work for sums going to ∞ , remember that a fraction to ∞ is just 0

There's more examples, but I'm too lazy to write them, so I'll just write the equations

Note

$$\sum_{k=0}^{\infty} kr^k = \sum_{k=1}^{\infty} kr^k = \boxed{\frac{r}{(1-r)^2}}$$

1.6 Section 2.6 - Matrices



Definition 22 (Directed Graph)

This is a **directed graph** and could be used to model:

- How webpages are linked
- Flights between cities
- Who knows who

How can a computer store this directed graph?

Definition 23 (Matrix)

We will use a **matrix**, which is a rectangular array of numbers

For this network, we define a matrix

$$A = [a_{ij}]$$

Where

$$a_{ij} = \begin{cases} 0 & \text{if web page } i \text{ doesn't link to } j \\ 1 & \text{if web page } i \text{ does link to } j \end{cases}$$

i = row (starts at 1)

j = column (starts at 1)

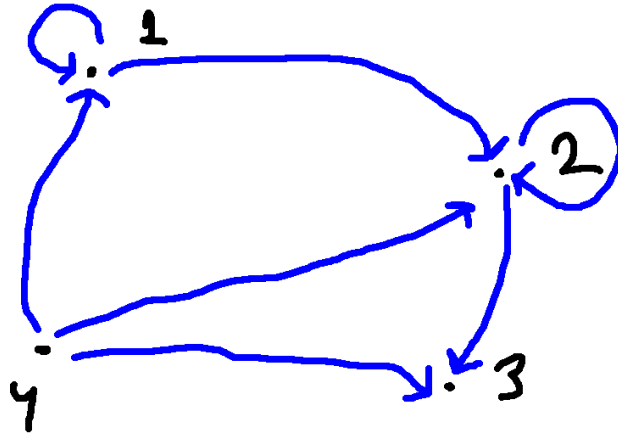
The resulting matrix would look like

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

For this network, the matrix is

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Let's modify the graph now



This matrix is

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Addition and Subtraction

We define the addition of matrices by adding corresponding components

$$A + B = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The same can be said for Subtraction

$$A - B = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Note

Two matrices can only be added/subtracted if they're the same size

Question

Going back to the first network, how many ways can we start at 1 and return to 1 after passing through exactly 1 webpage?

$$\begin{aligned}
 &1 \rightarrow 1 \rightarrow 1 \\
 &1 \rightarrow 2 \rightarrow 1 \\
 &a_{12} \rightarrow a_{21}
 \end{aligned}$$

$$\begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & 1 & 0 & 0 \\ a_{21} & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Example 13

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. $A + B$

Solution:

Undefined, due to dimensions

2. $6A$

Solution:

$$6A = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

3. AB

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ -1 & 11 \end{bmatrix}$$

4. BA

Solution:

$$BA = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

5. BC

Solution:

$$BC = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$$

6. CB

Solution:

Undefined, due to dimensions

7. $C^2 = CC$

$$C^2 = CC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} \\ = 5C$$

8. C^{11}

$$\begin{aligned}
C^2 &= 5C \\
C^3 &= C^2 \cdot C \\
&= 5C \cdot C \\
&= 5C^2 \\
&= 5^2C \\
C^{11} &= 5^{10}C = 5^{10} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}
\end{aligned}$$

Okay let's now define some properties

1.6.1 Properties of Matrices

Definition 24 (Additive Identity)

We define the **Additive Identity** for matrices as

$${}_{{n \times m}} A + \vec{0} = {}_{{n \times m}} A, \vec{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Definition 25 (Multiplicative Identity)

We define the **Multiplicative Identity** for matrices as

$${}_{{n \times m}} A \cdot I_n = A, I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

But A must be **square**!

Some matrices have a multiplicative inverse such that

Definition 26 (Multiplicative Inverse)

We define the **Multiplicative Inverse** of matrices as

$$A \cdot B = I$$

But only **some** matrices have them, not all do!

Also, A must be square to even qualify for possibly having one

For this class, we define

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 14

We have

$$\begin{bmatrix} 4 & 3 \\ 7 & 5 \end{bmatrix}^{-1} = \frac{1}{20 - 21} \begin{bmatrix} 5 & -3 \\ -7 & 4 \end{bmatrix}$$